Wireshark Manual

Task #2 — Network Protocol Analysis

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SECTION 1 — Purpose of Wireshark

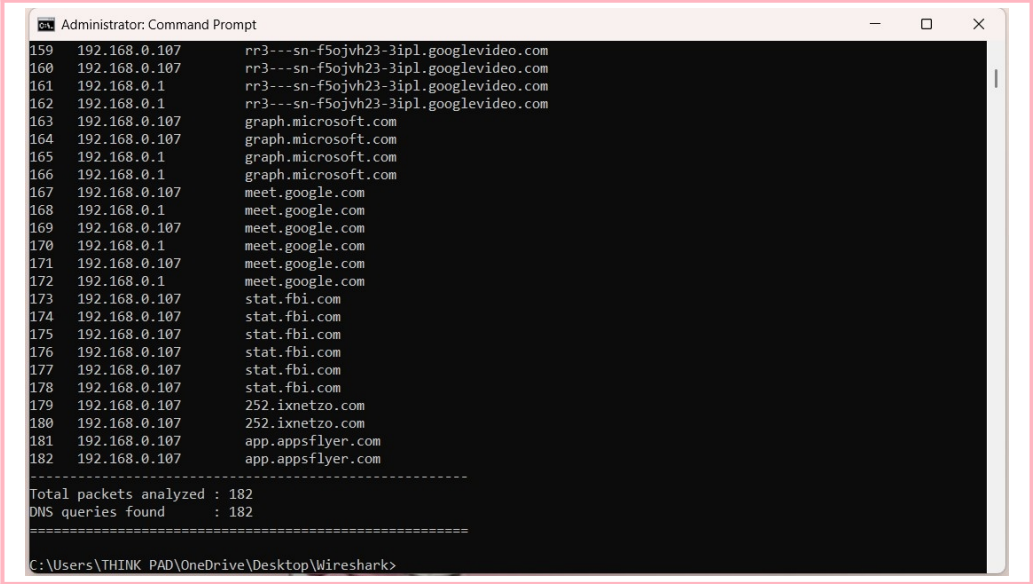
Purpose of Wireshark

Wireshark is a network protocol analyzer used to capture and analyze network traffic in real time. It allows users to inspect packets traveling across a network and examine the details of different protocols.

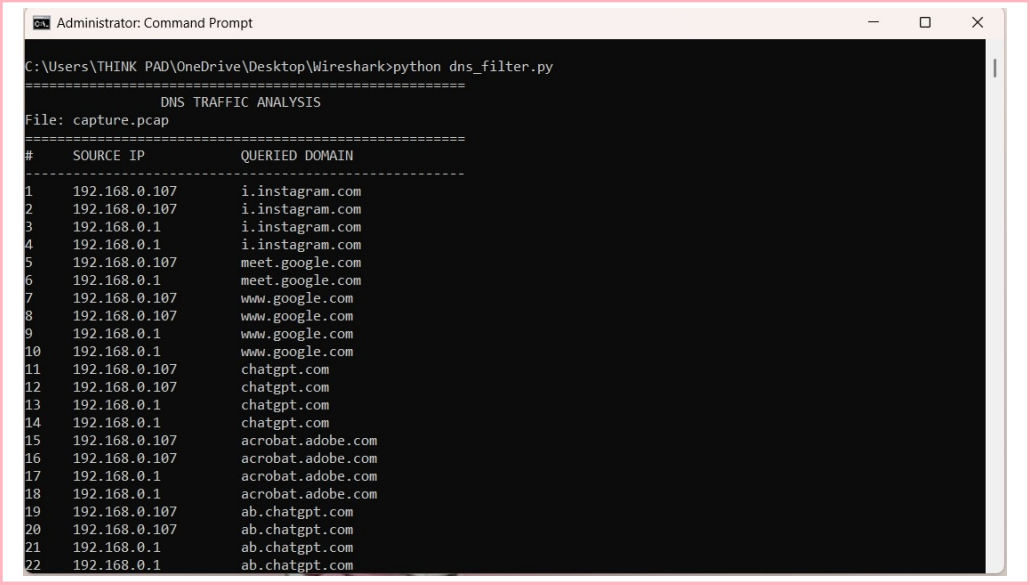
Wireshark is commonly used by network administrators, cybersecurity professionals, and researchers to troubleshoot network issues, monitor traffic, and detect suspicious activity.

Some common uses include:

- Monitoring network communication
- Troubleshooting connectivity problems
- Learning how network protocols work
- Detecting malicious or unusual traffic



Screenshot 1 — Wireshark Interface Overview



```
Administrator: Command Prompt

C:\Users\THINK PAD\OneDrive\Desktop\Wireshark>python dns_filter.py
=======================================================
                  DNS TRAFFIC ANALYSIS
File: capture.pcap
=======================================================
#     SOURCE IP            QUERIED DOMAIN
-------------------------------------------------------
1     192.168.0.107        i.instagram.com
2     192.168.0.107        i.instagram.com
3     192.168.0.1          i.instagram.com
4     192.168.0.1          i.instagram.com
5     192.168.0.107        meet.google.com
6     192.168.0.1          meet.google.com
7     192.168.0.107        www.google.com
8     192.168.0.107        www.google.com
9     192.168.0.1          www.google.com
10    192.168.0.1          www.google.com
11    192.168.0.107        chatgpt.com
12    192.168.0.107        chatgpt.com
13    192.168.0.1          chatgpt.com
14    192.168.0.1          chatgpt.com
15    192.168.0.107        acrobat.adobe.com
16    192.168.0.107        acrobat.adobe.com
17    192.168.0.1          acrobat.adobe.com
18    192.168.0.1          acrobat.adobe.com
19    192.168.0.107        ab.chatgpt.com
20    192.168.0.107        ab.chatgpt.com
21    192.168.0.1          ab.chatgpt.com
22    192.168.0.1          ab.chatgpt.com
```

Screenshot 2 — Packet Capture in Progress

SECTION 2 — How to Approach a Packet Capture

How to Approach a Packet Capture

When analyzing packet captures using Wireshark, a systematic approach is important.

1. Capture or Load Traffic

The first step is to capture network traffic from a network interface or open a previously saved packet capture file (.pcap).

2. Apply Filters

A packet capture may contain thousands of packets, so filters are used to narrow down the analysis.

3. Inspect Packet Details

Wireshark allows users to expand packets and view protocol layers such as Ethernet, IP, and application protocols.

4. Follow Network Conversations

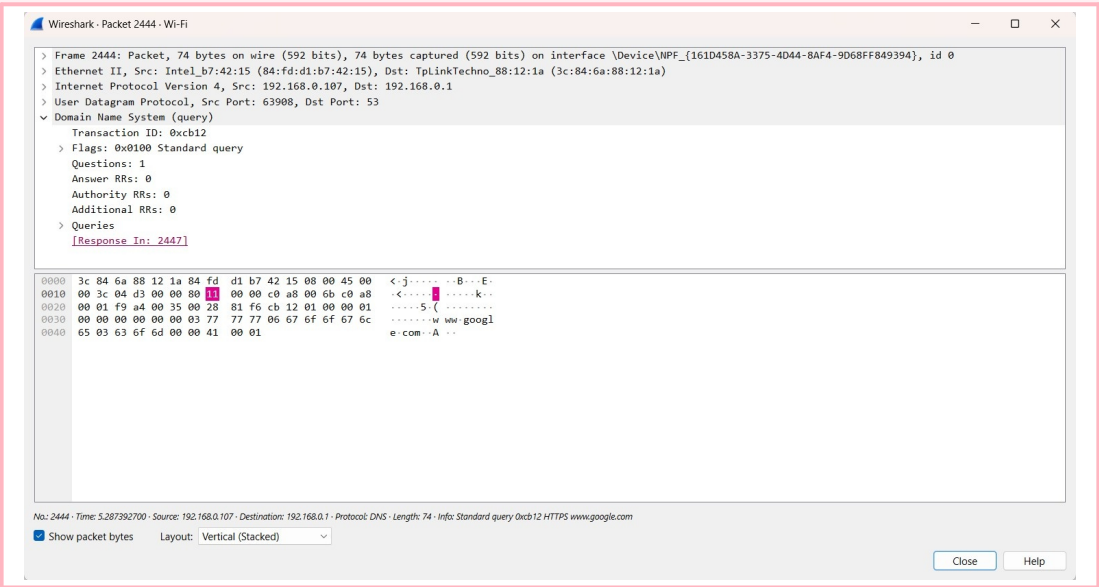
Users can follow streams to observe complete communications between devices.

5. Identify Suspicious Patterns

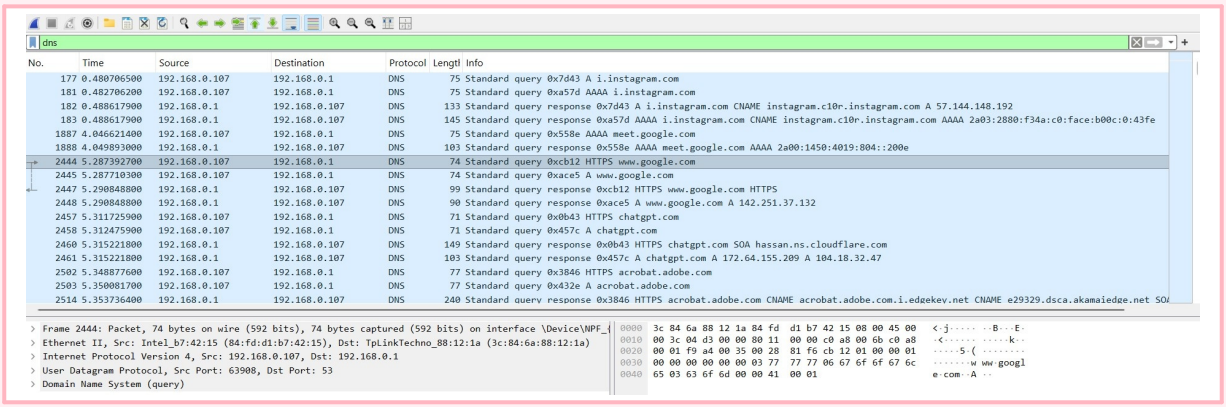
Look for unusual behavior such as repeated requests, unknown domains, or unexpected data transfers.

Example Filters

Filter	Description
`dns`	Show only DNS protocol packets
`http`	Show only HTTP traffic
`tcp`	Show all TCP packets



Screenshot 3 — Applying Display Filters in Wireshark



Screenshot 4 — Packet Details Pane Expanded

SECTION 3 — Protocol Deep Dive: DNS

DNS Protocol Overview

DNS (Domain Name System) is responsible for translating human-readable domain names into IP addresses. For example: **google.com → 142.250.190.78**

This allows users to access websites without remembering numerical IP addresses.

How DNS Works

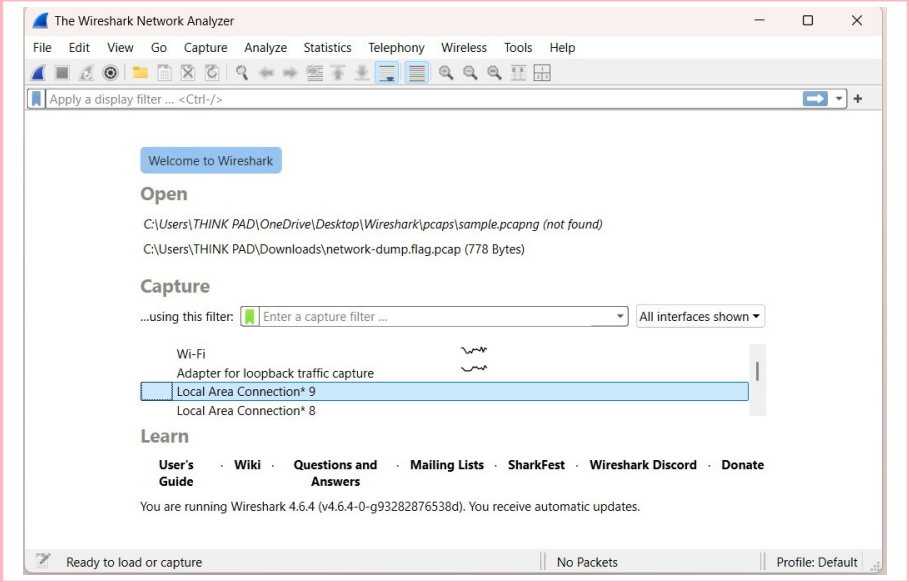
- A user enters a domain name into a browser.
- The system sends a DNS query to a DNS server.
- The DNS server searches its records.
- The server responds with the corresponding IP address.

DNS Fields in Wireshark

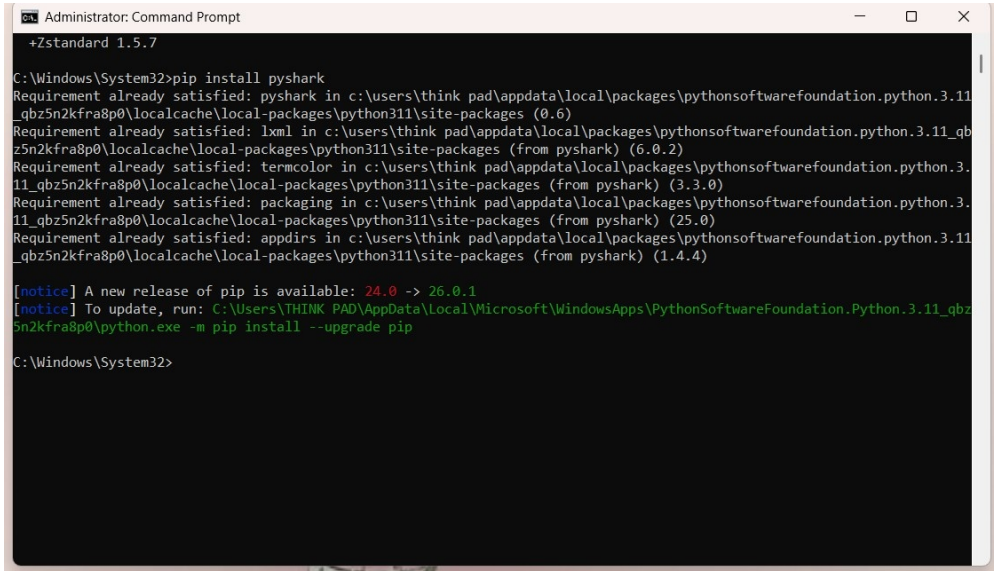
Field	Description
`Transaction ID`	Identifies DNS requests and responses
`Query Name`	The domain being requested
`Query Type`	Type of DNS record such as A or AAAA

DNS Filter in Wireshark

To view only DNS traffic, enter **dns** in the Wireshark display filter bar and press Enter.



Screenshot 5 — DNS Filter Applied in Wireshark



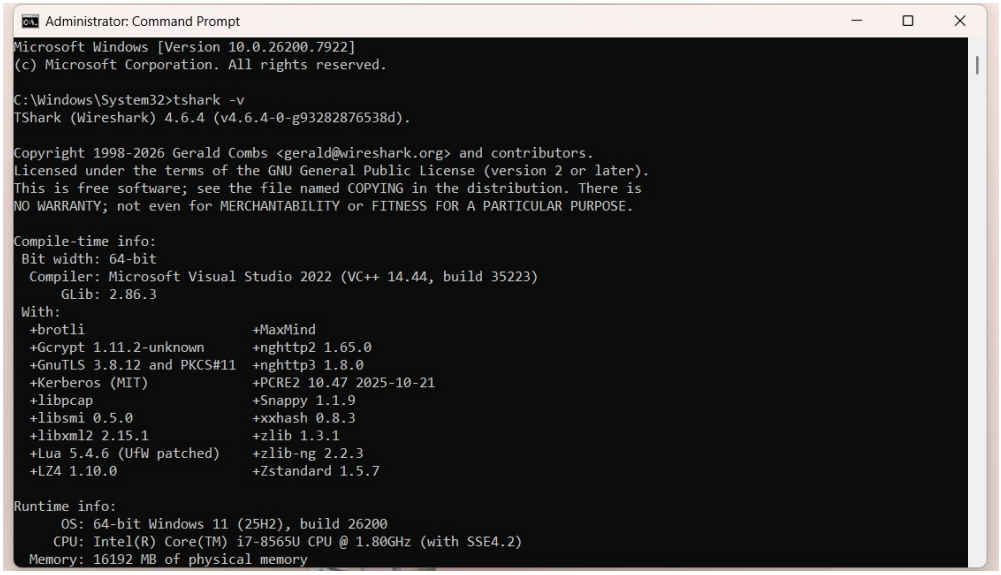
```
Administrator: Command Prompt
  +Zstandard 1.5.7

C:\Windows\System32>pip install pyshark
Requirement already satisfied: pyshark in c:\users\think pad\appdata\local\packages\pythonsoftwarefoundation.python.3.11
_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (0.6)
Requirement already satisfied: lxml in c:\users\think pad\appdata\local\packages\pythonsoftwarefoundation.python.3.11_qb
z5n2kfra8p0\localcache\local-packages\python311\site-packages (from pyshark) (6.0.2)
Requirement already satisfied: termcolor in c:\users\think pad\appdata\local\packages\pythonsoftwarefoundation.python.3.
11_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from pyshark) (3.3.0)
Requirement already satisfied: packaging in c:\users\think pad\appdata\local\packages\pythonsoftwarefoundation.python.3.
11_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from pyshark) (25.0)
Requirement already satisfied: appdirs in c:\users\think pad\appdata\local\packages\pythonsoftwarefoundation.python.3.11
_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from pyshark) (1.4.4)

[notice] A new release of pip is available: 24.0 -> 26.0.1
[notice] To update, run: C:\Users\THINK PAD\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.11_qbz
5n2kfra8p0\python.exe -m pip install --upgrade pip

C:\Windows\System32>
```

Screenshot 6 — DNS Query and Response Packets



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.7922]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>tshark -v
TShark (Wireshark) 4.6.4 (v4.6.4-0-g93282876538d).

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NO WARRANTY; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Compile-time info:
 Bit width: 64-bit
  Compiler: Microsoft Visual Studio 2022 (VC++ 14.44, build 35223)
      GLib: 2.86.3
 With:
  +brotli                      +MaxMind
  +Gcrypt 1.11.2-unknown       +nghttp2 1.65.0
  +GnuTLS 3.8.12 and PKCS#11   +nghttp3 1.8.0
  +Kerberos (MIT)              +PCRE2 10.47 2025-10-21
  +libpcap                     +Snappy 1.1.9
  +libsmi 0.5.0                +xxhash 0.8.3
  +libxml2 2.15.1              +zlib 1.3.1
  +Lua 5.4.6 (UfW patched)     +zlib-ng 2.2.3
  +LZ4 1.10.0                  +Zstandard 1.5.7

Runtime info:
      OS: 64-bit Windows 11 (25H2), build 26200
     CPU: Intel(R) Core(TM) i7-8565U CPU @ 1.80GHz (with SSE4.2)
  Memory: 16192 MB of physical memory
```

Screenshot 7 — DNS Transaction Details